

Molecular Characterization of β -Lactamase-Resistant *Klebsiella aerogenes* Isolated from Raw Milk in Bangladesh

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Abstract

The number of dairy farms in Bangladesh is steadily increasing, yet the characterization of common pathogenic organisms such as *Klebsiella aerogenes* remains underexplored. Due to the overwhelming challenges of antimicrobial resistance, the transmission of antibiotic resistance genes, and mobile colistin resistance (*mcr*) genes through zoonotic organisms, this study aimed to characterize *K. aerogenes* in raw milk in Sylhet. In this study, a total of 234 raw milk samples were collected from different dairy farms and milk-selling points in Sylhet. Then, isolates were detected by *16S rRNA* gene through PCR after standard cultural and biochemical tests. Furthermore, antibiotic sensitivity test was performed using the disc diffusion method, followed by screening of antibiotic-resistant genes (TEM, CTX-M, CTX-M-1, CTX-M-2a, SHV, OXA-1, CMY, and NDM-1) and *mcr* genes (*mcr*-1 to *mcr*-9). Finally, partial sequencing of *16S rRNA* was performed to study the genetic diversity. Among 234 samples, 86 (36.44%) were positive for *Klebsiella* and exhibited variation in resistance levels to amoxicillin (100%), ampicillin (100%), ceftazidime (100%), tetracycline (100%) followed by azithromycin (89.54%), cefoxitin (86.04%), gentamicin (83.74%), and cefuroxime (77.90%). All the samples harbored the *bla*_{TEM} gene (100%) and showed negative results when screening for *mcr* genes. The phylogenetic outcome reported that those isolates were closely related to *K. aerogenes*. According to our findings, the majority of the isolates were resistant to several antibiotics and contained the *bla*_{TEM} gene, which may represent a significant risk to the overall well-being of the population. These findings underscore that raw milk could be a potential harbor for multidrug resistance *K. aerogenes*, raising concerns about antimicrobial resistance transmission through traditionally processed dairy products, as such products are often prepared without proper heat treatment. The study emphasizes the need for strict hygiene practices and mandatory pasteurization measures during processing to reduce potential public health risks.

Keywords: raw milk, *Klebsiella aerogenes*, β -lactamase, antimicrobial resistance, Bangladesh

Introduction

In recent decades, antimicrobial resistance (AMR) has become a global challenge due to the rapid emergence of antimicrobial-resistant bacterial strains, particularly *Klebsiella spp.*, which is a gram-negative organism responsible for causing mastitis, and their potential harm to the dairy sector is a major concern, including in Bangladesh (Massé et al., 2020; Nahar et al., 2022; Oliveira et al., 2013; Salauddin et al., 2019). The primary way that *Klebsiella spp.* are spread in dairy cattle is by contact with the teats, mostly from animal manure, bedding, and other farm equipment (Cheng et al.,

2020). Nevertheless, in the last 10 years or so, there has been a significant global increase in the incidence of *Klebsiella spp.* found in milk samples (Song et al., 2023).

Among the species of *Klebsiella*, *Klebsiella aerogenes* is a significant pathogen related to infections obtained in hospital settings. *K. aerogenes*, which is widely regarded as an opportunistic pathogen, is linked to nosocomial epidemics and is responsible for infections of the urinary system, reproductive tract, skin and soft tissue, and circulation (Davin-Regli and Pagès, 2015). Although this organism is not often present in milk or other dairy products, it can invade the milk value chain through the environment or due to improper

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handling practiced by milk processors (Owusu-Kwarteng et al., 2020).

An increasing number of instances have reported the emergence of multidrug resistant (MDR) *Klebsiella* that produce extended-spectrum β -lactamase enzymes, such as CTX-M, SHV, and TEM. The mobile colistin resistance (*mcr*) genes have been documented globally in *Enterobacteriales*, including *Escherichia coli*, *Salmonella*, and *K. pneumoniae*, originating from various sources, particularly the environment and animals (Elbediwi et al., 2019). Recent reports from the World Health Organization indicate that the misuse of antibiotics in food-producing animals plays a significant role in the emergence of AMR, which can spread to humans through environmental pathways, direct contact, or the food chain (World Health Organization, 2015).

All these factors together raise concerns and focus on the importance of this study. A few studies have been conducted on this matter in Bangladesh (Nahar et al., 2022; Salauddin et al., 2019). Hence, this study intended to investigate the prevalence, antibiotic resistance, and *mcr* profiling of *Klebsiella* spp. (depending on the *16S rRNA* gene) in raw milk collected from dairy farms and milk-selling points in different areas of Sylhet Sadar, Bangladesh. The research also focuses on the phylogenetic analysis depending on the *16S rRNA* gene of MDR, antibiotic resistance gene (ARG) gene and *mcr* gene carrying positive isolates from raw milk.

Instead of evaluating zoonotic transmission directly, the study focuses on the dynamics of contamination at the farm level and the potential role of raw milk as a reservoir of MDR *K. aerogenes*. By characterizing *K. aerogenes* in the raw milk samples, the study aims to provide valuable region-based data for improving food safety practices and mitigating the risks associated with *K. aerogenes* contamination in Bangladesh.

Materials and Methods

Phenotypic and genotypic characterization of *Klebsiella* spp.

Experimental scheme and sampling. A total of 234 raw milk samples were collected from randomly selected different dairy farms and selling points across nine areas of Sylhet Sadar (Tilagor, Alurtol, Shahporan, Jalalabad, Bateshwar, Akhali, Pathantula, Alompur, and Kazirbazar) in the Sylhet district of Bangladesh. Before collecting the raw milk samples, sterilization of all the equipment was conducted. The milk from each of the four udders was subsequently transferred directly into a sterile 15 mL falcon tube and transported via icebox to the lab for microbiological analysis. Before the analysis started, samples were maintained in a refrigerator (around 4°C).

Isolation of *Klebsiella* spp. About 10 mL of each sample was homogenized with distilled water and diluted with the same diluent. Then, the samples were pre-enriched by culturing in the nutrient broth (Oxoid Ltd, United Kingdom) in a ratio of 1:10 while kept in incubation overnight at 37°C. The eosin methylene blue (EMB) agar (Oxoid Ltd, United Kingdom) was used to isolate *Klebsiella* spp. colonies. In the presence of *Klebsiella* spp., the medium produces a large mucoid colony with a pink tint. Bacterial colonies of pink or purple were considered *Klebsiella* colonies primarily.

Repeated culture of these presumptive colonies was performed to obtain a pure single colony. Then, these colonies were subjected to Gram staining and biochemical tests.

Molecular detection of *Klebsiella* spp. through the 16S rRNA gene. The universal gene (*16S rRNA*) for *Klebsiella* spp. was targeted for molecular confirmation following the methods described earlier by Ibrahim et al. (2019) with little modifications (Supplementary Table S3). DNA was extracted from pure culture using the conventional boiling method (Chowdhury et al., 2025; Rana et al., 2024; Roy et al., 2024). Electrophoresis of the amplified product was done at 60 V for 45 min in 2% agarose gel. Ethidium bromide was used for staining, and the gel was examined through a UV-UV-transilluminator using a 100 bp DNA ladder.

Antibiotic sensitivity test

The antibiotic sensitivity test (AST) of isolated *Klebsiella* spp. was carried out by following the disc diffusion method according to CLSI guideline 2020 (Bauer et al., 1966). A group of commonly used antibiotics (Supplementary Table S1) was tested against all the isolated positive samples by following protocols with some modifications (Ferdaus et al., 2019). The CLSI guidelines were followed in the interpretation of the data (resistant, intermediate, and sensitive) (Supplementary Table S2). An isolate showing resistance against three or more types of antibiotics was considered MDR in nature (Sweeney et al., 2018).

Genotypic detection of ARGs and *mcr*

The genetic basis underlying phenotypic resistance was screened through PCR with particular primers for antibiotic-resistance genes. Genes responsible for resistance to β -lactam and carbapenems were amplified (Supplementary Table S4), according to amplification conditions described by the references. Moreover, we screened those positive samples for the presence of any mobile colistin-resistant gene (*mcr-1* to *mcr-9*; Supplementary Table S5).

Phylogenetic analysis

PCR products of *16S rRNA* were sequenced. A total of 41 sequences were analyzed using the BLAST query tool (<http://blast.ncbi.nlm.nih.gov>) to determine nonchance sequence similarities. BLAST queries were restricted to the 16S ribosomal RNA database (Ibrahim et al., 2019). Alignment of the 41 sequences was carried out using ClustalW. Following the utilization of the sequences obtained, the phylogenetic tree was constructed employing MEGA 11.0 software. The tree was constructed using the neighbor-joining algorithm. The distances were calculated using the method known as maximum composite likelihood (Nei and Saitou, 1987).

Statistical analysis

The present study's data were entered into Excel 365 (Microsoft/Office 365, Redmond, DC, USA) and then transferred to GraphPad Prism (Prism 8.4.2, San Diego, CA, USA). Graphs were generated using the statistics software GraphPad Prism 9.3.1 and R-studio.

The prevalence % was calculated using the following formula:

$$\text{Prevalence \%} = \frac{\text{Number of cases in a population at a point of time}}{\text{Total population at the same point of time}} \times 100$$

Results

Results of cultural, morphological, and biochemical examination of *Klebsiella* spp.

Presumptive identification of *Klebsiella* spp. was based on colony morphology on EMB agar, Gram staining, and standard biochemical tests. All isolates were indole-negative, MR-negative, VP-positive, citrate-positive, urease-positive, and catalase-positive (Supplementary Fig. S1 and Fig. S2).

The percentage of presumed *Klebsiella* spp. on EMB agar was found in 95 samples (40.25%) out of 234 raw milk samples. The biochemical test (+ve) *Klebsiella* spp. samples were found to be 89 samples (37.71%) out of the 236 raw milk samples. Moreover, out of 95 culture-positive samples, around 93.68% ($n = 89$) of biochemical-positive *Klebsiella* spp. samples were detected (Supplementary Fig. S2).

Molecular detection and prevalence of *Klebsiella* spp.

Out of the 234 raw milk samples tested, 86 of them, which account for 36.44% of the total, were determined to be positive for *Klebsiella* spp. in the molecular detection process (Fig. 1A). In Jalalabad, Sylhet, the prevalence of *Klebsiella* spp. isolates was determined to be 23.07% (6 out of 26 samples), which was the lowest among all locations (Fig. 1B). The Kazirbazar area exhibited the greatest prevalence rate of 42.3% (11 out of 26 samples) according to Figure 1B. Furthermore, among the 89 positive *Klebsiella* spp. samples from biochemical testing (as shown in Supplementary Fig. S2), PCR analysis (as depicted in Fig. 1) verified 86 samples,

accounting for 96.62% of the total. The PCR-based molecular detection of *Klebsiella* spp. resulted in the visualization of amplified DNA, which exhibited a required band size of 1419 bp, as observed in Figure 1.

Antibiotic resistance pattern

Antimicrobial susceptibility test by disc diffusion method revealed that those isolates were fully resistant against amoxicillin (100%), ampicillin (100%), ceftazidime (100%), tetracycline (100%) followed by azithromycin (89.54%), cefoxitin (86.04%), gentamicin (83.74%), and cefuroxime (77.90%; Fig. 3 and Supplementary Fig. S3). Those isolates were mostly sensitive to trimethoprim-sulfamethoxazole (93.02%), followed by nalidixic acid (86.04%), aztreonam (81.25%), imipenem (73.25%), meropenem (68.75%), and ticarcillin (55.81%; Fig. 2 and Supplementary Table S6).

Moreover, a total of 100% ($n = 86$ out of 86) isolates were found to be MDR (resistant against at least one antimicrobial in equal or more than four antimicrobial classes). However, no specific pattern was observed. Moreover, ($n = 3$) showed resistance against ≥ 8 antibiotic classes, ($n = 9$) showed resistance against ≥ 7 antibiotics classes, ($n = 24$) resistant against ≥ 6 , ($n = 40$) resistant against ≥ 5 , ($n = 10$) resistant against ≥ 4 , and ($n = 0$) resistant against ≥ 3 class of antibiotics (Fig. 3 and Supplementary Table S7).

Phenotypic resistance profile

The isolates exhibited the highest resistance to CXM, AK, TE, CAZ, AMX, AMP, CTX, S, GEN, AZM, and CX. They demonstrated intermediate sensitivity to TIC, NOR, CIP, and LEV. Conversely, they were mostly sensitive to SXT, AT, NA, MEM, and IPM. Overall resistance patterns are plotted in the heatmap (Fig. 4).

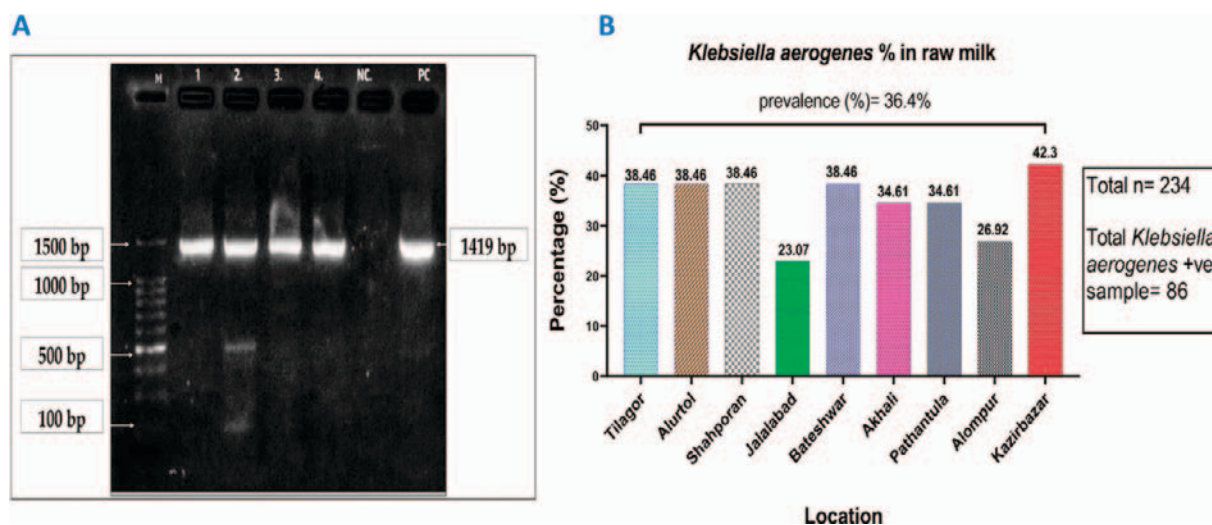


FIG. 1. Detection and prevalence of *Klebsiella* spp. in raw milk samples. (A) Molecular detection of amplified DNA of 16S rRNA gene of *Klebsiella* spp. isolates. Lane M: 100 bp DNA ladder; lanes 1–4: Positive samples showing the expected 1419 bp amplicon. (B) Prevalence of *Klebsiella* spp. in raw milk in different dairy farms and selling point of Sylhet Sadar. The percentage of positive samples is indicated above each bar. The overall prevalence of *Klebsiella* spp. across all locations was 36.4% (86/234 samples). NC, negative control.

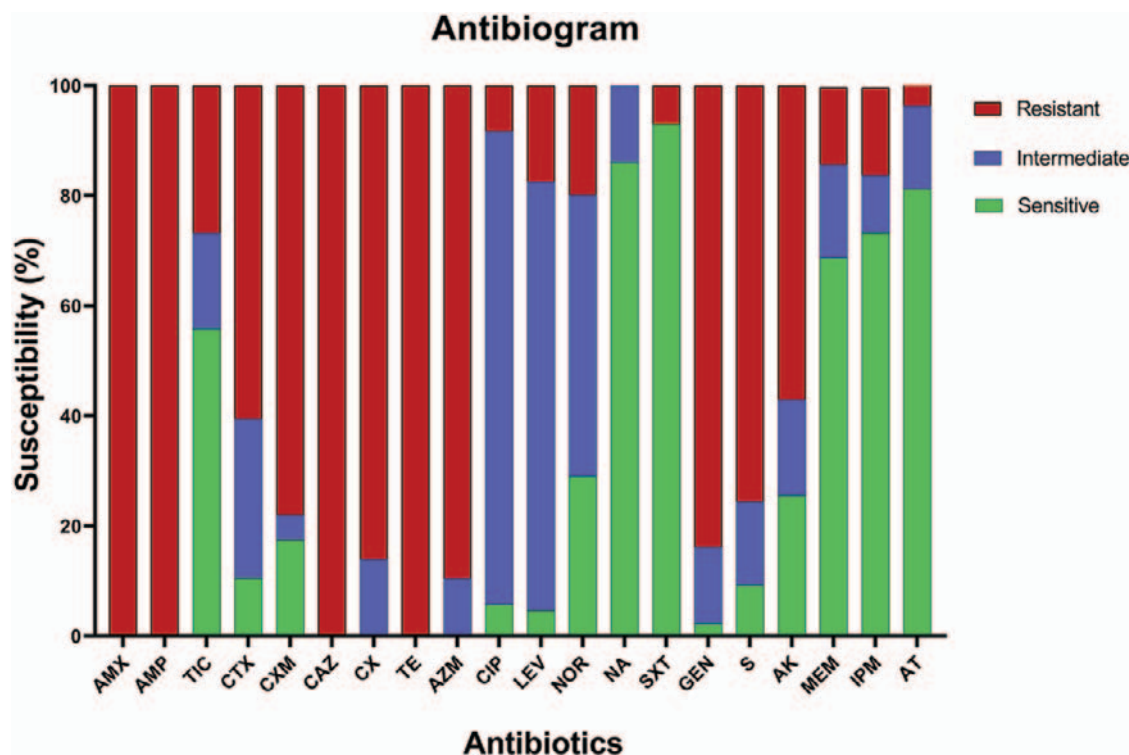


FIG. 2. Heatmap showing the overall antibiogram profile of *Klebsiella* spp. positive isolates. CXM, cefuroxime; AK, amikacin; TE, tetracycline; CAZ, ceftazidime; AMX, amoxicillin; AMP, ampicillin; S, streptomycin; GEN, gentamicin; AZM, azithromycin; TIC, ticarcillin; NOR, norfloxacin; CIP, ciprofloxacin; LEV, levofloxacin; SXT, trimethoprim-sulfamethoxazole; AT, aztreonam; NA, nalidixic acid; MEM, meropenem; IPM, imipenem.

Molecular detection of ARGs and *mcr* genes in isolated *Klebsiella* spp.

A total of 86 positive *Klebsiella* spp. isolates were further screened for any ARGs containing the gene and found that all samples harbored the *bla*_{TEM} gene (100%). However, other ARG genes were absent in our study (Fig. 5).

Conversely, our isolates tested negative for *mcr* genes (*mcr-1* to *mcr-9*) in PCR analysis (Supplementary Fig. S3).

Phylogenetic analysis of MDR *Klebsiella* spp. isolate

Phylogenetic analysis of the amplified DNA sequence of 16S rRNA gene of MDR *Klebsiella* spp. revealed that our

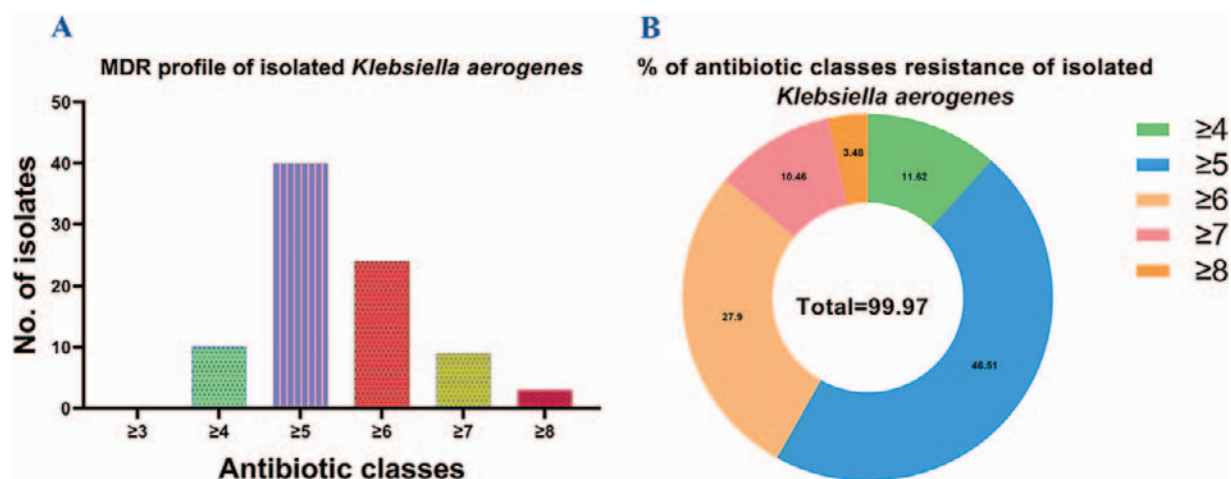


FIG. 3. Multidrug resistance (MDR) profile of isolated *Klebsiella* spp. (A) depicts no. of isolates that showed MDR characteristics against antibiotic classes; (B) illustrates the percentage of isolates that were resistant against multiple antibiotic classes or groups; antibiotic classes or groups are classified as ≥3 (3 or < 3 antibiotic classes), ≥4 (4 or < 4 antibiotic classes), ≥5 (5 or < 5 antibiotic classes), ≥6 (6 or < 6 antibiotic classes), ≥7 (7 or < 7 antibiotic classes), and ≥8 (8 or < 8 antibiotic classes).

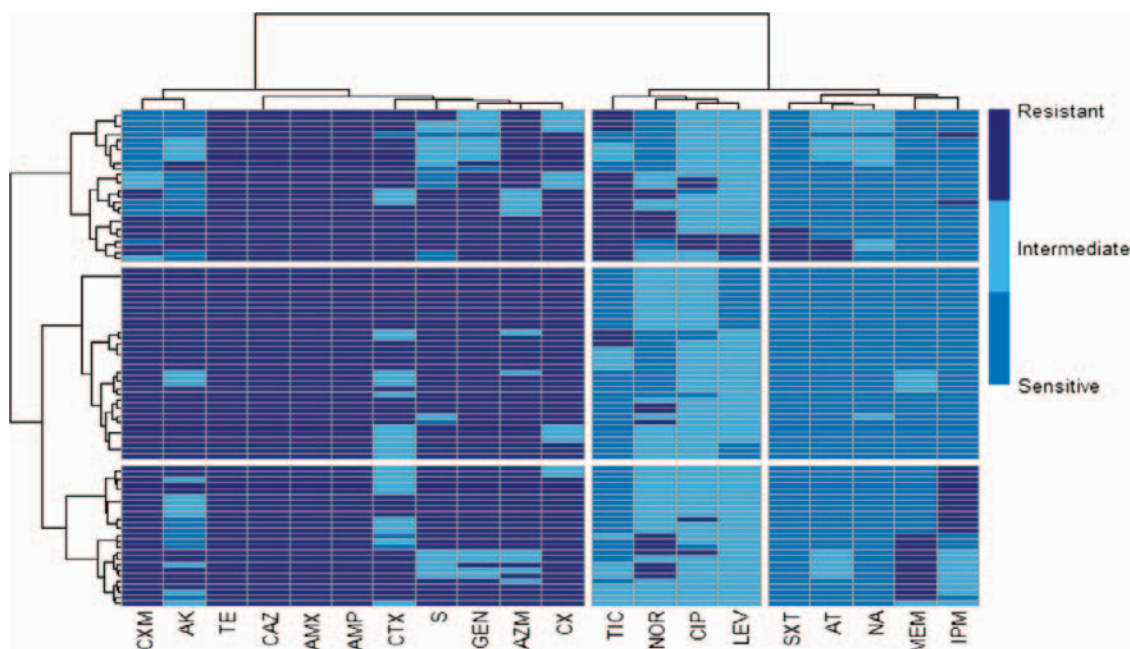


FIG. 4. Heatmap showing phenotypic resistance profile with clustering of antibiotics and *Klebsiella spp.* isolates. All 86 isolates were assessed against 20 antimicrobial agents. The color dodgerblue3 = sensitive, steelblue1 = intermediate, and navy = resistant. CXM, cefuroxime; AK, amikacin; TE, tetracycline; CAZ, ceftazidime; AMX, amoxicillin; AMP, ampicillin; S, streptomycin; GEN, gentamicin; AZM, azithromycin; TIC, ticarcillin; NOR, norfloxacin; CIP, ciprofloxacin; LEV, levofloxacin; SXT, trimethoprim-sulfamethoxazole; AT, aztreonam; NA, nalidixic acid; MEM, meropenem; IPM, imipenem.

study sequence was closely related to KY621520.1 *K. aerogenes* strain OP3 and OQ405498.1 *K. aerogenes* strain C29 (branch distance = 0.0031). Furthermore, several other strains were closely related to our study sequence (Fig. 6). In terms of similarity, PP702222.1 *K. aerogenes* strain EA144 and PP702219.1 *K. aerogenes* strain EA154 were the furthest from our study sequences (Fig. 6). So that the isolated *Klebsiella spp.* was confirmed as *K. aerogenes* after performing partial sequencing of *16S rRNA* gene.

Discussions

The findings of this study demonstrate the prevalence and antibiotic resistance patterns of *K. aerogenes* isolated from raw milk in Bangladesh. Our observations indicate that the *Klebsiella* species is commonly found in raw milk. In this study, the prevalence of *Klebsiella spp.* through morphological confirmation was 40.25%, whereas the prevalence of molecular identification was 36.44%. One possible explanation for the discrepancy in prevalence between molecular confirmation

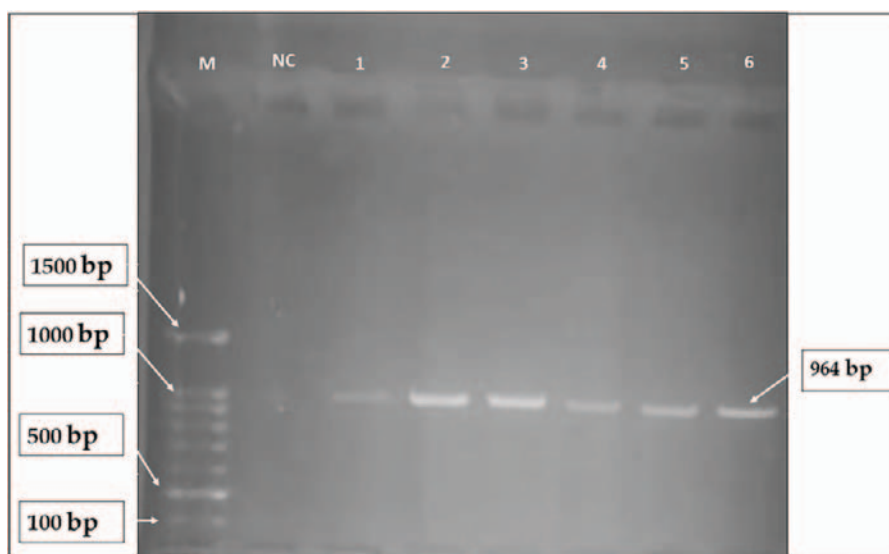


FIG. 5. Amplified DNA of *blaTEM* gene. Antibiotic resistance gene of isolated *Klebsiella spp.* at 964 bp. (lane M: 100 bp DNA ladder; Lane NC: (-ve) control; lane 1–6: *blaTEM* positive isolates).

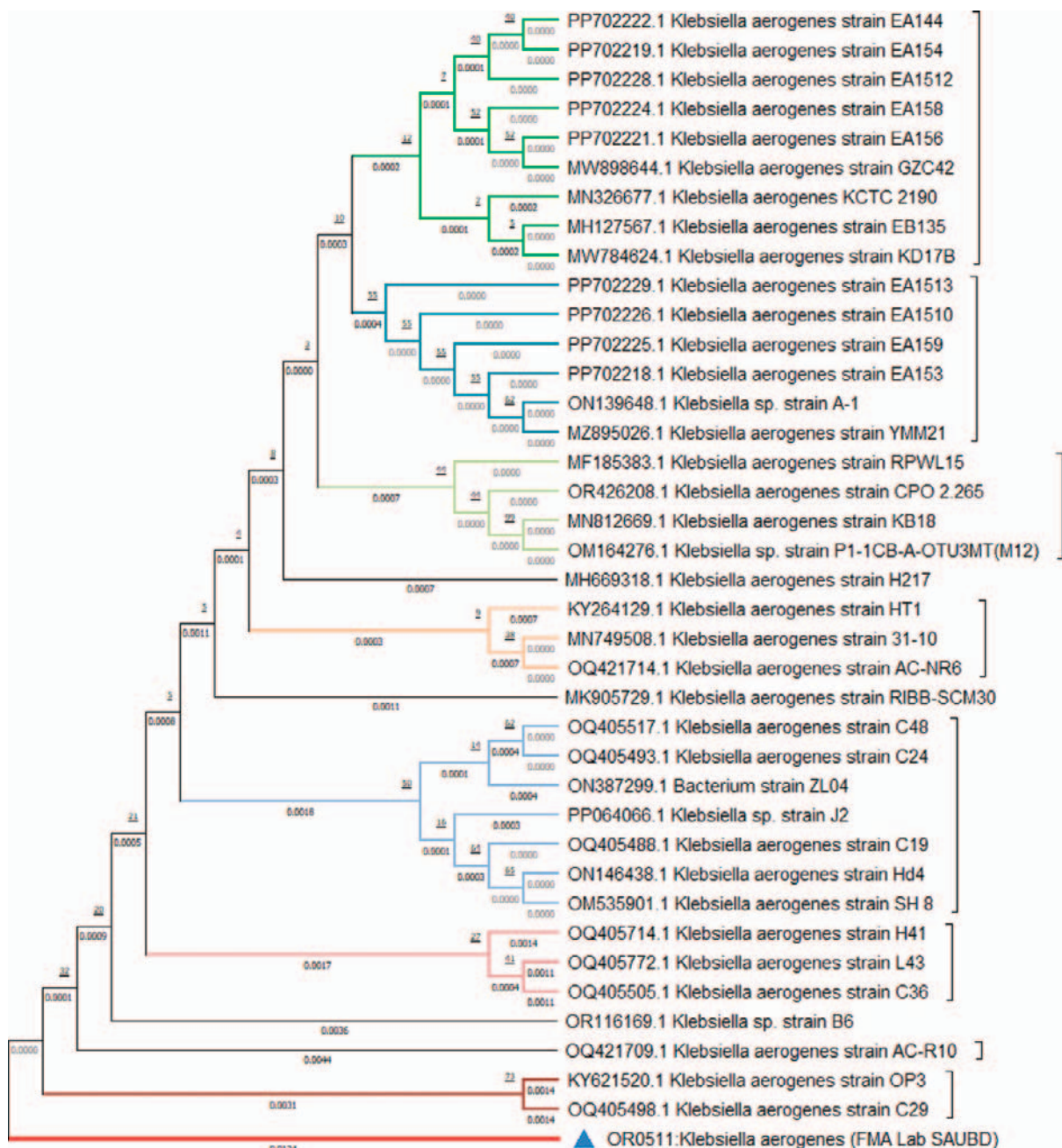


FIG. 6. Phylogenetic analysis of amplified DNA of 16 s rRNA gene of *Klebsiella spp.* isolated from raw milk sample. Phylogenetic tree based on the 16 s rRNA gene sequence from this study and those from other reference strains, constructed by the maximum-likelihood method using MEGA software, version X (<http://www.megasoftware.net/>). The evolutionary history was inferred using the neighbor-joining method. The optimal tree is shown (next to the branches). The evolutionary distances were computed using the maximum composite likelihood method and are in the units of the number of base substitutions per site. All ambiguous positions were removed for each sequence pair (pairwise deletion option).

(36.44%) and culture-based identification (40.25%) is that EMB agar is nonspecific, allowing the growth of various lactose-fermenting, mucoid colonies that resemble *Klebsiella spp.* (Osman et al., 2020).

This issue is further intensified by the widespread antibiotic resistance observed in these isolates, highlighting the critical need for enhanced surveillance and management strategies to reduce these risks. The high prevalence of *Klebsiella* contamination, observed in raw milk samples in this study (Fig. 1), highlights the persistent food safety challenges in the milk value chain.

Similar studies in Bangladesh, India, and Sri Lanka reported *Klebsiella spp.* prevalence in raw milk ranging from 30% to 54.8%. A study previously conducted in Mymensingh, Bangladesh, which corroborated our findings, discovered that 30% (30/100) of fresh milk samples were contaminated with *Klebsiella spp.* (Nahar et al., 2022). A separate investigation carried out in Jessore, Bangladesh, similarly revealed a notable occurrence of *Klebsiella spp.* in unpasteurized milk, surpassing the safety threshold index advised by the European Committee (Tasnim and Islam, 2015). A similar trend was

also reported in India, where they reported the prevalence was 27% (27/100; Chauhan et al., 2013). Sri Lanka, a neighboring country of Bangladesh, showed a significantly greater proportion of *Klebsiella* spp. in their milk samples, with a prevalence rate of 54.8% (17/31; Jayaweera et al., 2018), which is higher than ours. In contrast, Uttar Pradesh, situated in the eastern region of India, detected a comparatively lower proportion (7.92%) of this microorganism out of 240 fresh milk samples (Yadav et al., 2021).

A critical finding in this study was the extensive MDR exhibited by our isolates (Fig. 2 and Fig. 3). If we do not halt the mushrooming pandemic of antibiotic resistance now, it could cause problems on a global scale. Resistance to several antibiotics has been observed in *Klebsiella* spp. isolated from raw milk samples, which highlights the rising threat of AMR in the milk value chain. These findings are consistent with previous studies that have reported similar resistance patterns in *Klebsiella* spp. isolated from raw milk (Enferad and Mahdavi, 2021; Gundogan and Yakar, 2007; Nahar et al., 2022; Nobrega et al., 2021; Tasnim and Islam, 2015; Yadav et al., 2021). The resistance may result from constant antibiotic pressure, the acquisition of antibiotics from environmental sources, or farm laborers.

In contrast to other studied classes, the majority of isolates showed susceptibility to carbapenem and monobactam antibiotics. Nahar and the team observed that those isolates from Mymensingh showed 100% susceptibility to imipenem (Nahar et al., 2022). Although we found some of our isolates were resistant to this antibiotic, the percentage was surprisingly lower (15.95%) than other antibiotics. The study conducted by Al-Zahrani and Akhtar from Saudi Arabia indicated that carbapenem, namely the imipenem derivative, was the most effective antibiotic. This was supported by the fact that 96.29% of isolates were sensitive to carbapenems (Al-Zahrani and Akhtar, 2005). This fact is contrary to the study, which was conducted in Sudan and observed a relatively higher resistance % to this antibiotic (49.4%; Badri et al., 2017). The discrepancy in the findings of the antibiogram test across multiple studies might likely be attributed to variations in geographical location, treatment protocols, and the measurement of antibiotic usage in different countries (Enferad and Mahdavi, 2021). Numerous studies also found similar MDR patterns in *Klebsiella* spp. isolates in milk (Badri et al., 2017; Nahar et al., 2022; Salauddin et al., 2019; Tasnim and Islam, 2015; Yang et al., 2021).

In our study, all positive *Klebsiella* spp. isolates were found to carry the *bla*_{TEM} gene (Fig. 5), highlighting a concerning trend of resistance. These findings are consistent with studies from other regions reporting a higher prevalence of the *bla*_{TEM} gene (Enferad and Mahdavi, 2021; Tekiner and Özpınar, 2016). A group of researchers in India also noted the presence of the *bla*_{TEM} gene in bovine milk (Koovapra et al., 2016). In Pakistan, mastitic cow milk was also found to have a greater percentage (56%) of the *bla*_{TEM} gene (Jamal et al., 2024). Nevertheless, the absence of the *mcr* gene offers a glimmer of hope for our region.

We studied the phylogeny of our isolates and found that this organism is closely linked with *K. aerogenes* (Fig. 6). A study in Indonesia also claimed the presence of *K. aerogenes* in pindang (a food made from tuna fish; Rachmawati et al., 2022). In addition, a recent study carried out in India

released the results of their work, in which the research team also identified the presence of this organism in street food (namely, fried rice). This identification was validated using partial sequencing of the *16S rRNA* gene (Parivendhan M et al., 2024). Many researchers have already confirmed that using *16S rRNA* sequence analysis offers a more accurate and objective way to identify foodborne bacteria (Chukwu et al., 2019).

A major drawback of our study was that we did not screen the human samples from those intimately associated with the farms or milk-selling areas. By doing a screening of those samples, it may be feasible to ascertain the precise sequence of events that led to the contamination of milk samples by this bacterium. Again, this study was limited to the Sylhet region; no research was done in other parts of Bangladesh. Furthermore, whole genome sequencing or plasmid profiling is a must to understand whether it is plasmid or chromosome borne to explore the real mechanism behind horizontal transmission. However, this limitation does not reduce the importance and reliability of the current study findings. Furthermore, we think our results are comprehensive enough to explain the current situation of MDR *K. aerogenes* in raw milk and will assist legislators in implementing legislation on hygienic procedures in the dairy industry.

Conclusion

The current study was intended to characterize the *K. aerogenes* in raw milk from different dairy farms and milk-selling points of Sylhet Sadar, and those isolates were found to be highly prevalent in raw milk. Moreover, those isolates were MDR and were resistant to at least four or more groups of antibiotics. Those positive isolates were found to be 100% resistant to ampicillin, amoxicillin, ceftazidime, and tetracycline. However, in our study, we found that those isolates were highly sensitive to monobactams (aztreonam: 81.25%) and carbapenem (imipenem: 73.25%, meropenem: 68.75%). All the samples were capable of harboring the *bla*_{TEM} gene (100%). The outcome of the phylogenetic study showed that our isolates were closely related to *K. aerogenes*, which aligned with other findings from different parts of the world. This outcome suggests that the raw milk of cows may get contaminated by the environment, or the animal itself may be acting as a contamination source. This outcome also suggests that the raw milk may serve as a potential source of spreading zoonotic MDR-*K. aerogenes* to humans, which is an alarming issue. The implementation of suitable control and preventative measures and the development of long-term strategies, such as establishing regional AMR surveillance within dairy supply chains, implementing farmer education programs focused on antibiotic stewardship, and promoting mandatory milk pasteurization, are necessary to ensure food safety and reduce the resistance transmission. It is necessary to conduct additional studies on MDR-*Klebsiella* species, with a special emphasis on the genomic diversity of these bacteria and antibiotic alternatives, to lessen the amount of MDR-*Klebsiella* that is present in milk.

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Authors' Contributions

F.J.F.: Writing—original draft, methodology, investigation, and formal analysis. A.A.P.: Methodology, investigation, and data curation. S.R.: Investigation and data curation. S.U.: Software and methodology. M.N.: Resources and project administration. S.U.M.: Software, investigation, and formal analysis. K.Z.: Validation, project administration, and conceptualization. A.A.: Methodology and project administration. F.M.A.H.: Writing—review and editing, supervision, and conceptualization.

Availability of Data and Materials

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Author Disclosure Statement

No competing financial interests exist.

Funding Information

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Supplementary Material

Supplementary Data

References

- Al-Zahrani AJ, Akhtar N. Susceptibility patterns of extended spectrum SS-Lactamase (ESBL)-producing *Escherichia coli* and *Klebsiella pneumoniae* isolated in a teaching hospital. *Pak J Med Res* 2005;44(2).
- Badri AM, Ibrahim IT, Mohamed SG, et al. Prevalence of Extended Spectrum Beta Lactamase (ESBL) producing *Escherichia coli* and *Klebsiella pneumoniae* isolated from raw milk samples in Al Jazirah State, Sudan. *Mol Biol* 2017;7(1); doi: 10.4172/2168-9547.1000201
- Bauer AW, Kirby WMM, Sherris JC, et al. Antibiotic susceptibility testing by a standardized single disk method. *Am J Clin Pathol* 1966;45(4):493–496; doi: 10.1093/ajcp/45.4_ts.493
- Chauhan S, Farooq U, Singh V, et al. Determination Of Prevalence And Antibacterial Activity Of ESBL (Extended Spectrum Beta-Lactamases) Producing *Klebsiella* Species Isolated From Raw Milk Of Doon Valley In India. 2013.
- Cheng J, Zhang J, Han B, et al. *Klebsiella pneumoniae* isolated from bovine mastitis is cytopathogenic for bovine mammary epithelial cells. *J Dairy Sci* 2020;103(4):3493–3504; doi: 10.3168/jds.2019-17458
- Chowdhury T, Roy MC, Hossain FMA. Prevalence and zoonotic risk of multidrug-resistant *Escherichia Coli* in bovine subclinical mastitis milk: Insights into the virulence and antimicrobial resistance. *Food Sci Nutr* 2025;13(1):e4761; doi: 10.1002/fsn3.4761
- Chukwu EE, Nwaokorie FO, Coker AO, et al. 16S rRNA gene sequencing: A practical approach to confirming the identity of food borne bacteria. *Ife J Sci* 2019;21(3):13; doi: 10.4314/ijfs.v21i3.2
- Davin-Regli A, Pagès J-M. Enterobacter aerogenes and enterobacter cloacae; versatile bacterial pathogens confronting antibiotic treatment. *Front Microbiol* 2015;6:392; doi: 10.3389/fmicb.2015.00392
- Elbediwi M, Li Y, Paudyal N, et al. Global burden of colistin-resistant bacteria: Mobilized colistin resistance genes study (1980–2018). *Microorganisms* 2019;7(10):461; doi: 10.3390/microorganisms7100461
- Enferad E, Mahdavi S. Antibiotic resistance pattern and frequency of some beta lactamase genes in *Klebsiella pneumoniae* isolated from raw milk samples in Iran. *J Hellenic Vet Med Soc* 2021;71(4):2455; doi: 10.12681/jhvms.25925
- Ferdous H, Anjuman A, Mahfujur R, et al. Characterization of bacterial isolates, antibiogram profile and pro-inflammatory cytokines in subclinical mastitis in cross-bred dairy cows. *AJVS* 2019;62(2):1; doi: 10.5455/ajvs.58885
- Gundogan N, Yakar UA. Siderophore production, serum resistance, hemolytic activity and extended-spectrum β -lactamase-producing *Klebsiella* species isolated from milk and milk products. *J Food Safety* 2007;27(3):251–264; doi: 10.1111/j.1745-4565.2007.00077.x
- Ibrahim I, Kareem TA, Azeez YM, et al. Phylogenetic tree analysis based on the 16S sequence alignment for *Klebsiella* Spp. isolated from different sources. *Iraqi J Sci* 2019;2618–2628; doi: 10.24996/ijfs.2019.60.12.10
- Jamal M, Rahman SU, Khan M, et al. Genomic diversity and nutritional analysis of multi-drug resistant extended spectrum β -Lactamase producing-Klebsiella Pneumoniae genes isolated from mastitic cattle milk in district Peshawar, Pakistan. *Heliyon* 2024;10(15):e35876; doi: 10.1016/j.heliyon.2024.e35876
- Jayaweera SPJT, Randika LPCJ, Gamage HGCL, et al. Isolation of escherichia coli, Klebsiella Spp. and Staphylococcus Spp. from bovine mastitic milk in Nuwera Eliya District of Sri Lanka and their sensitivity to commonly used antibiotics. *South Asian J Res Microbiol* 2018;1–13; doi: 10.9734/sajrm/2018/v1i2766
- Koovapra S, Bandyopadhyay S, Das G, et al. Molecular signature of extended spectrum β -Lactamase producing Klebsiella pneumoniae isolated from bovine milk in Eastern and North-Eastern India. *Infect Genet Evol* 2016;44:395–402; doi: 10.1016/j.meegid.2016.07.032
- Massé J, Dufour S, Archambault M. Characterization of Klebsiella isolates obtained from clinical mastitis cases in dairy cattle. *J Dairy Sci* 2020;103(4):3392–3400; doi: 10.3168/jds.2019-17324
- Nahar A, Islam AKMA, Islam MN, et al. ESBL-producing *Klebsiella* Sp. isolated from raw milk of healthy cow in small holder dairy farms of Mymensingh District in Bangladesh. *BJVM* 2022;19(2); doi: 10.33109/bjvmjd2021amrt1
- Nei M, Saitou N. The neighbor-joining method: A new method for reconstructing phylogenetic trees. *Mol Biol Evol* 1987; 4(4):406–425; doi: 10.1093/oxfordjournals.molbev.a040454
- Nobrega DB, Calarga AP, Nascimento LC, et al. Molecular characterization of antimicrobial resistance in *Klebsiella pneumoniae* isolated from Brazilian Dairy Herds. *J Dairy Sci* 2021;104(6):7210–7224; doi: 10.3168/jds.2020-19569
- Oliveira L, Hulland C, Ruegg PL. Characterization of clinical mastitis occurring in cows on 50 large dairy herds in Wisconsin. *J Dairy Sci* 2013;96(12):7538–7549; doi: 10.3168/jds.2012-6078
- Osman EA, El-Amin N, Adrees EAE, et al. Comparing conventional, biochemical and genotypic methods for accurate

- identification of *Klebsiella Pneumoniae* in Sudan. Access Microbiol 2020;2(3):acmi000096; doi: 10.1099/acmi.0.000096
- Owusu-Kwarteng J, Akabanda F, Agyei D, et al. Microbial safety of milk production and fermented dairy products in Africa. Microorganisms 2020;8(5):752; doi: 10.3390/microorganisms8050752
- Parivendhan M, Harini M, Dharaniya A, et al. Molecular identification of bacteria (*Klebsiella Aerogenes*, PP335235) isolated from street food (Fried Rice); Study of its antibiotic sensitivity pattern and probiotic fortification of fruit juice (*Fragaria Ananassa*, *Passiflora Ligularis* and *Phyllanthus* 2024.
- Rachmawati N, Powell SM, Triwibowo R, et al. Modelling growth and histamine formation of *Klebsiella aerogenes* TI24 isolated from Indonesian Pindang. Int J Food Microbiol 2022;362:109459; doi: 10.1016/j.ijfoodmicro.2021.109459
- Rana S, Fahim F, Das R, et al. High prevalence of multidrug-resistant extended spectrum Beta Lactamase -Producing *Escherichia Coli* in raw milk in Bangladesh. Microbes and Infectious Diseases 2024;0(0):0–0; doi: 10.21608/mid.2024.258318.1735
- Roy MC, Chowdhury T, Hossain MT, et al. Zoonotic linkage and environmental contamination of Methicillin-Resistant *Staphylococcus Aureus* (MRSA) in dairy farms: A one health perspective. One Health 2024;18:100680; doi: 10.1016/j.onehlt.2024.100680
- Salaudhin M, Akter M, Hossain M, et al. Isolation of multi-drug resistant *Klebsiella* Sp. from bovine mastitis samples in Rangpur, Bangladesh. J Adv Vet Anim Res 2019;6(3):362–365; doi: 10.5455/javar.2019.f355
- Song J, Xiang W, Wang Q, et al. Prevalence and risk factors of *Klebsiella* Spp. in milk samples from dairy cows with mastitis—A global systematic review. Front Vet Sci 2023;10:1143257; doi: 10.3389/fvets.2023.1143257
- Sweeney MT, Lubbers BV, Schwarz S, et al. Applying definitions for multidrug resistance, extensive drug resistance and pandrug resistance to clinically significant livestock and companion animal bacterial pathogens. J Antimicrob Chemother 2018;73(6):1460–1463; doi: 10.1093/jac/dky043
- Tasnim U, Islam M. Pathogenic and drug resistant bacteria in raw milk of Jessore City: A potential food safety threat. Bangl J Vet Med 2015;13(1):71–78; doi: 10.3329/bjvm.v13i1.23723
- Tekiner İH, Özpınar H. Occurrence and characteristics of extended spectrum Beta-Lactamases-Producing Enterobacteriaceae from foods of animal origin. Braz J Microbiol 2016; 47(2):444–451; doi: 10.1016/j.bjm.2015.11.034
- World Health Organization. Global Action Plan on Antimicrobial Resistance. World Health Organization: Geneva; 2015.
- Yadav V, Joshi RK, Joshi N, et al. Detection of Carbapenemase Genes (*Bla*-NDM, *Bla*-KPC, *Bla*-OXA-48) in *Escherichia Coli* and *Klebsiella species*, isolated from milk samples of Bovine in Eastern Plain Zone of Uttar Pradesh. Indian J Anim Res 2021; doi: 10.18805/IJAR.B-4543
- Yang Y, Peng Y, Jiang J, et al. Isolation and characterization of multidrug-resistant *Klebsiella Pneumoniae* from raw cow milk in Jiangsu and Shandong Provinces, China. Transbound Emerg Dis 2021;68(3):1033–1039; doi: 10.1111/tbed.13787

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